VLAN Principles



Foreword

A Virtual Local Area Network (VLAN) represents a form of administrative network that defines a logical grouping of hosts or end system devices that are not limited to a physical location, and may be defined based on a wide range of parameters that allow for a greater flexibility in the way that logical groups are defined. The application of VLAN technology has expanded to support many aspects of enterprise networking as a means of logical data flow management and isolation.

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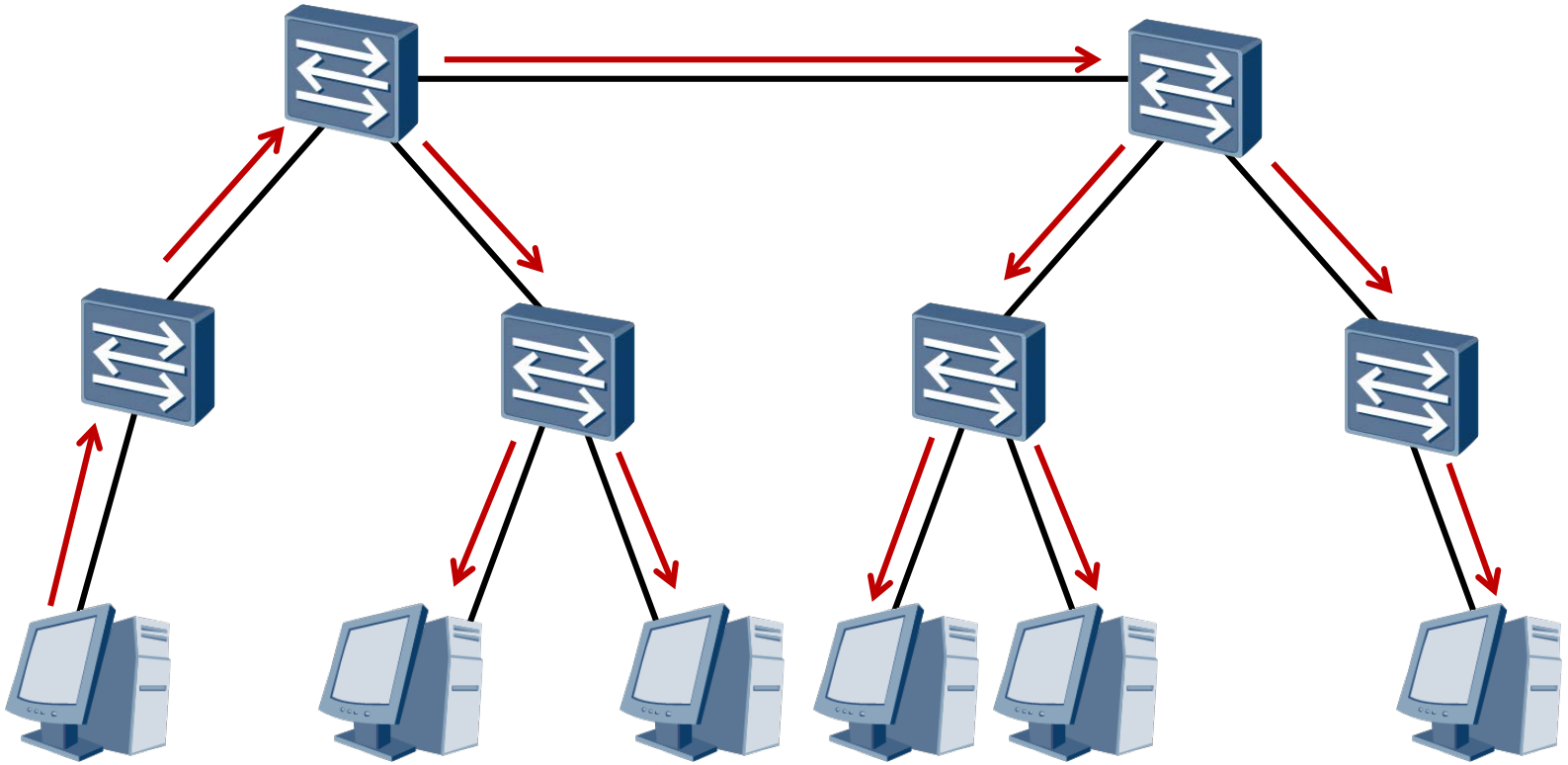
Objectives

Upon completion of this section, trainees will be able to:

- Explain the application of VLAN tagging.
- Describe the different port link types and characteristics.
- Successfully establish port based VLANs.

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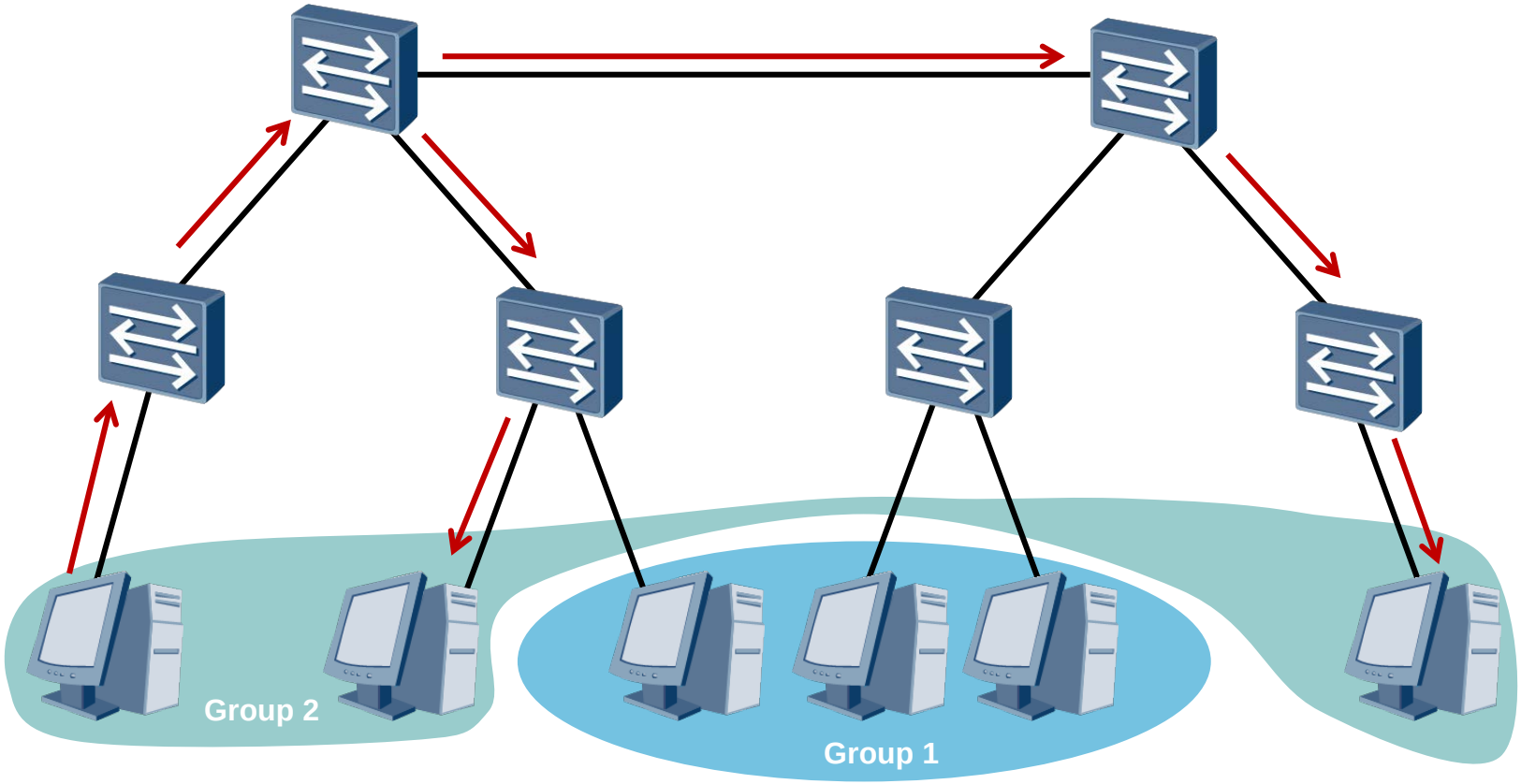
LAN Limitations



- No broadcast domain to manage expanding local networks.

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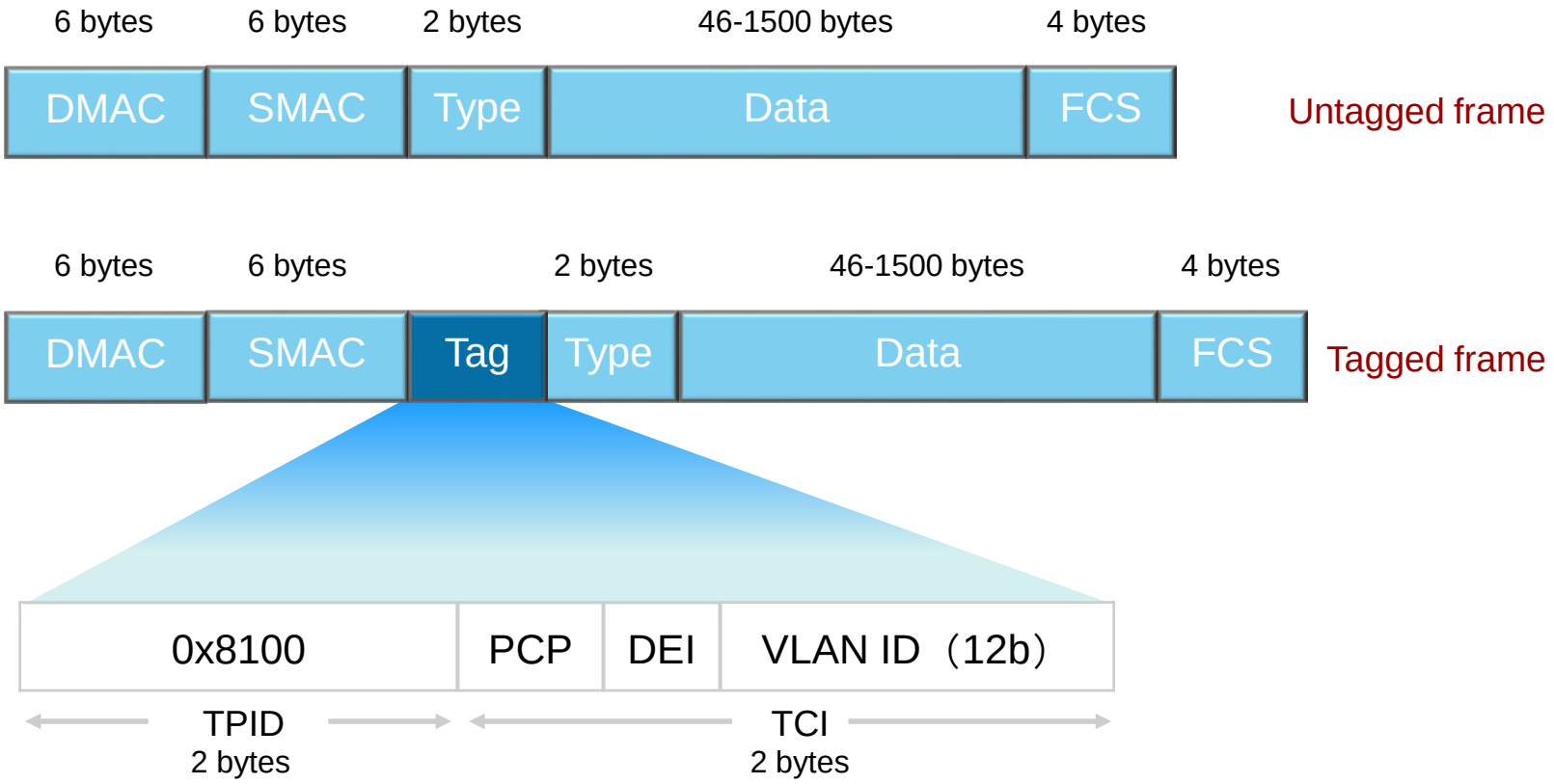
VLAN Technology



- A VLAN enables logical isolation of traffic at the data link layer.

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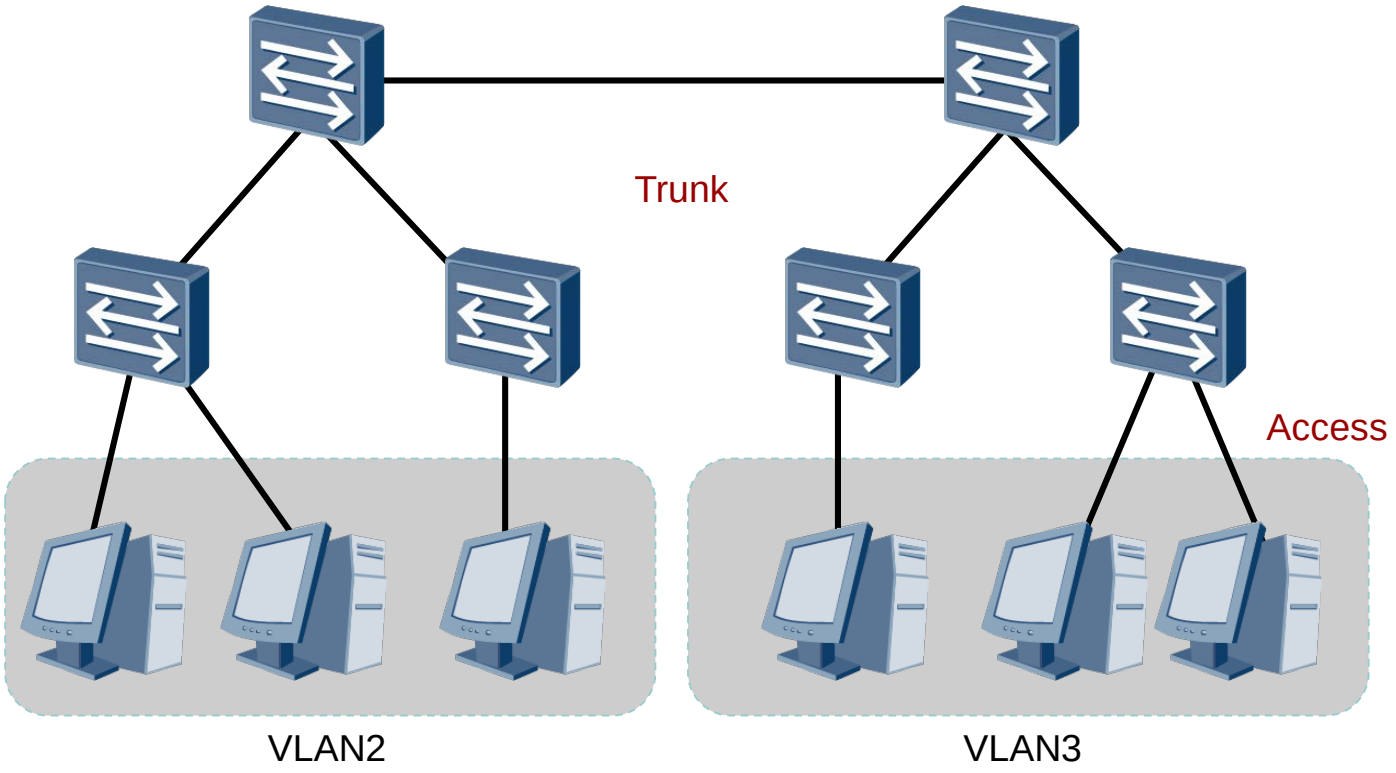
VLAN Frame Format



- A VLAN tag is inserted to distinguish frames for each VLAN.

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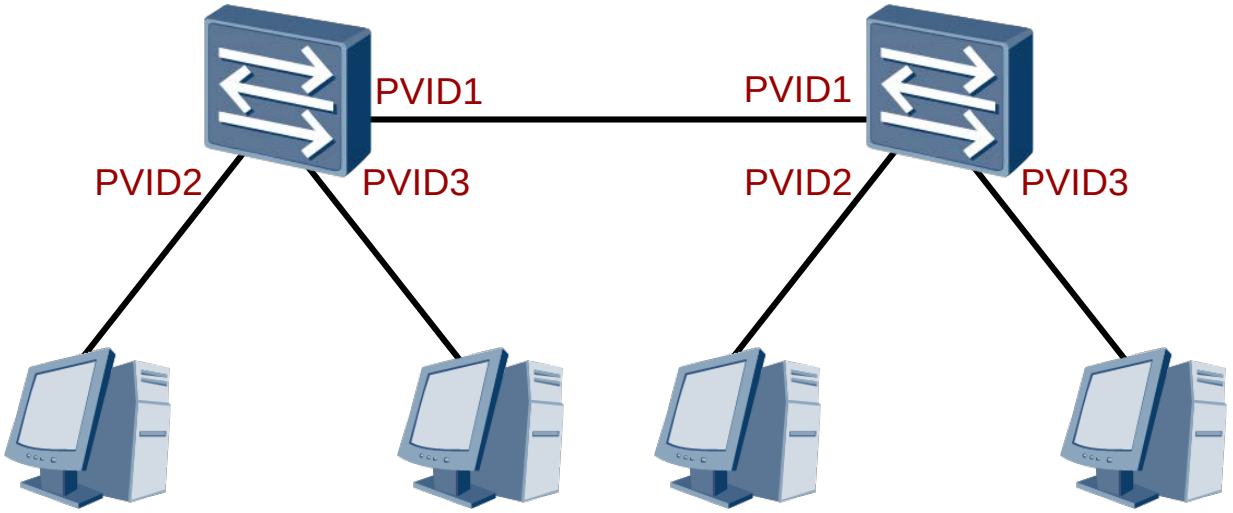
Link Types



- A trunk represents a backbone for the transmission of VLAN traffic between switches.

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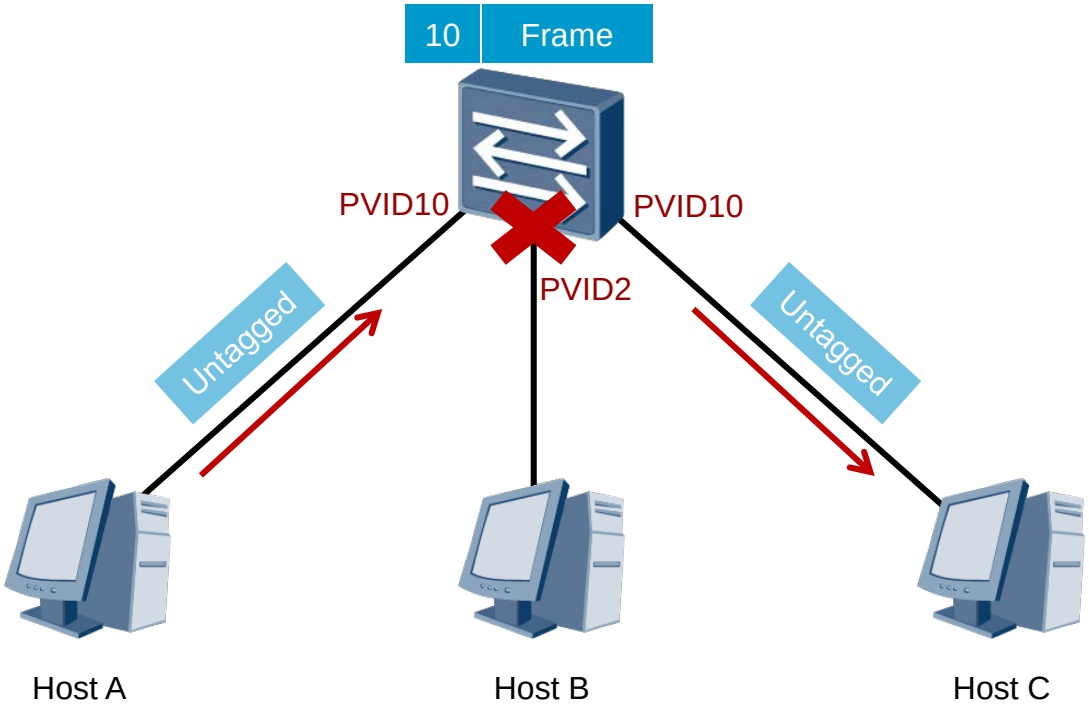
Port VLAN ID



- PVID represents the default VLAN for each interface.
- The PVID is set to VLAN 1 for all ports by default.

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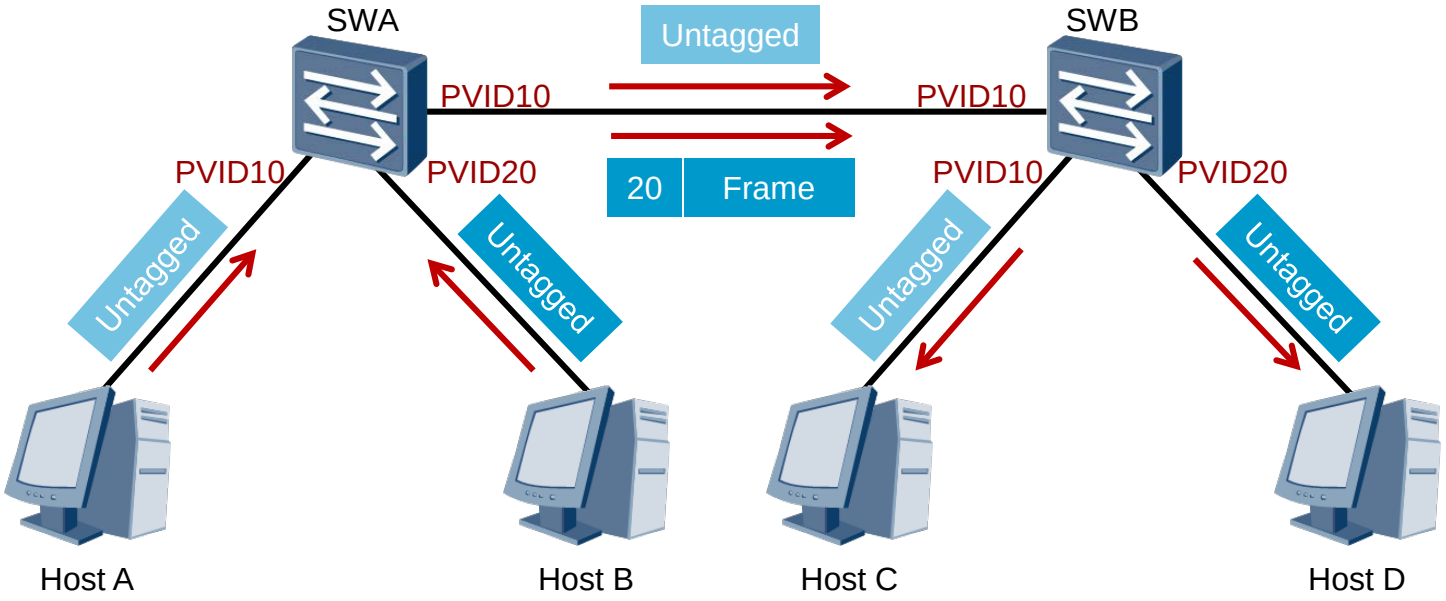
Port Types – Access



- Access ports remove VLAN tags before forwarding frames.

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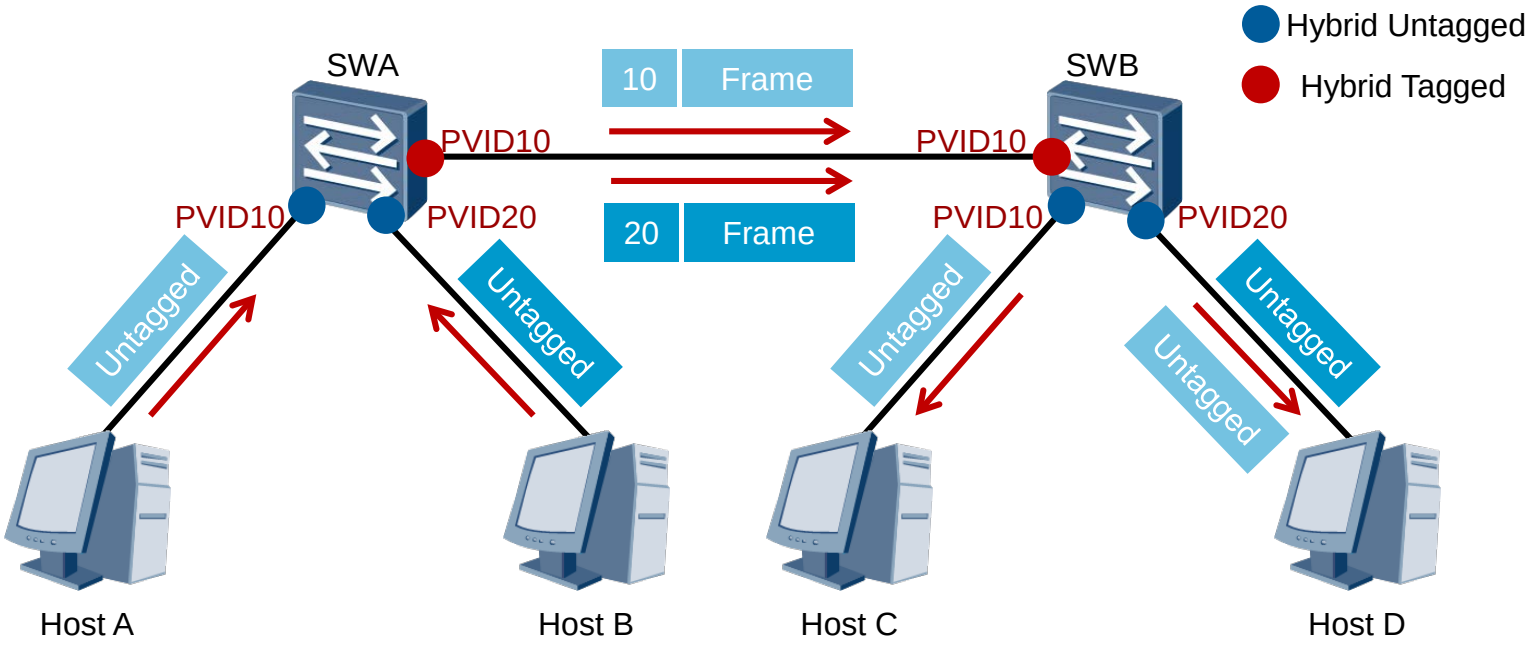
Port Types – Trunk



- Frames carried over a trunk link may be tagged or untagged.
- All VLANs must be permitted before being carried over a trunk.

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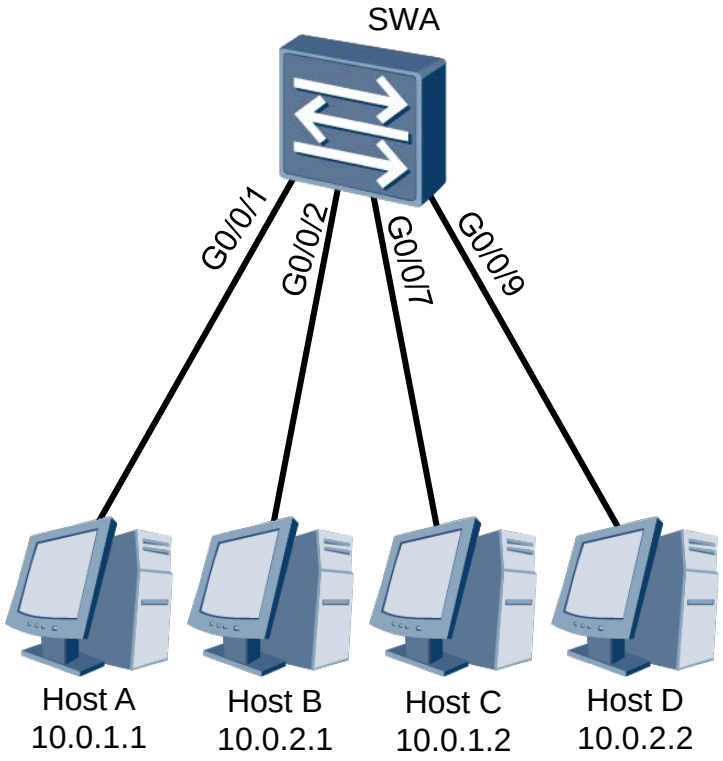
Port Types-Hybrid



- Hybrid ports are defined as either tagged or untagged
- VLAN communication can be managed on a port by port basis.

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VLAN Assignment Methods

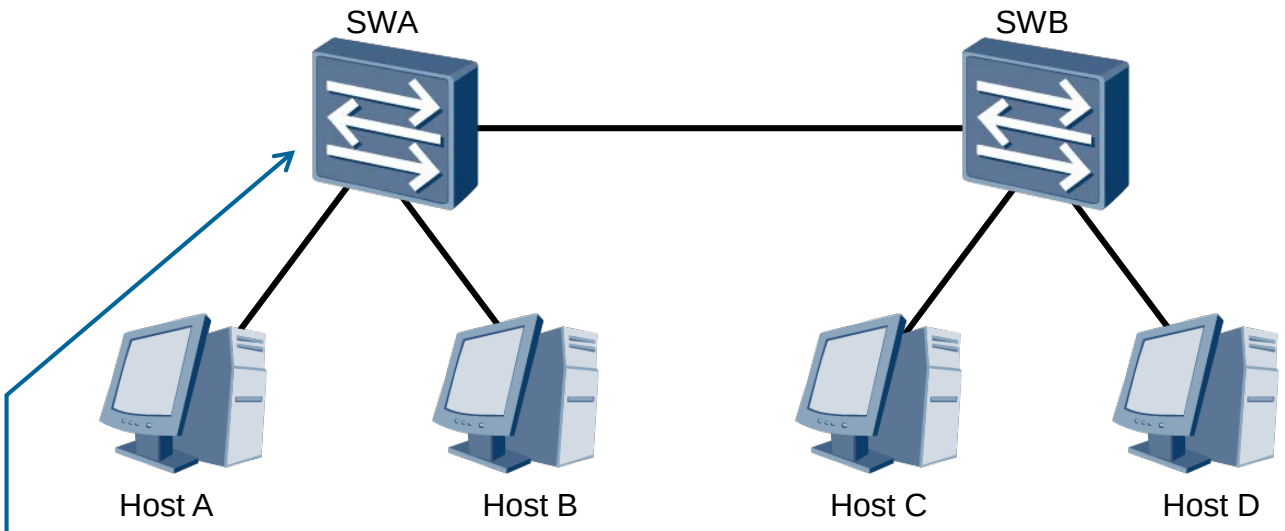


Assignment Method	VLAN 5	VLAN 10
Port based	G0/0/1, G0/0/7	G0/0/2 G0/0/9
MAC based	00-01-02-03-04-AA 00-01-02-03-04-CC	00-01-02-03-04-BB 00-01-02-03-04-DD
IP Subnet based	10.0.1.*	10.0.2.*
Protocol based	IP	IPX
Policy based	10.0.1.* + G0/0/1+ 00-01-02-03-04-AA	10.0.2.* + G0/0/2 + 00-01-02-03-04-BB

- Five methods of VLAN assignment are possible.
- Port based VLAN assignment is the default assignment method.

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Creating VLANs




```
[SWA]vlan 10
[SWA-vlan10]quit
[SWA]vlan batch 2 to 3
Info: This operation may take a few seconds. Please wait for a
moment...done.
```

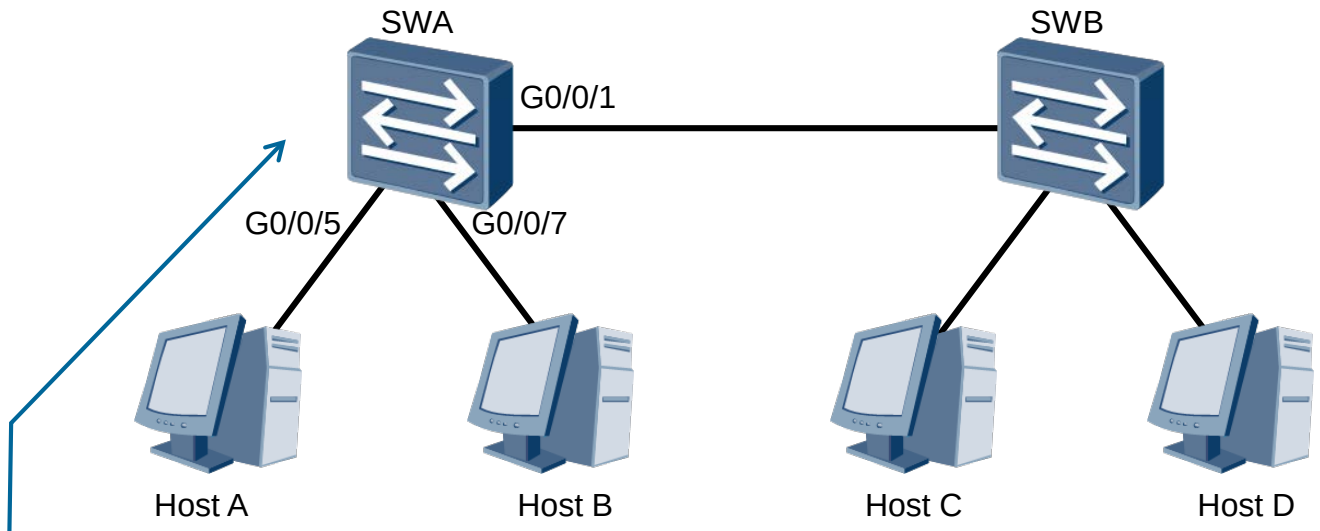
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Creating VLANs

```
[SWA]display vlan
The total number of vlans is : 4
--------------------------------------------------------------
U:Up; D:Down; TG:Tagged; UT:Untagged; MP:Vlan-mapping;
ST:Vlan-stacking; #: ProtocolTransparent-vlan; *:Management-
vlan;
--------------------------------------------------------------
VID  Type    Ports
--------------------------------------------------------------
1    common  UT:GE0/0/1(U) ……
2    common
3    common
10   common
……
```

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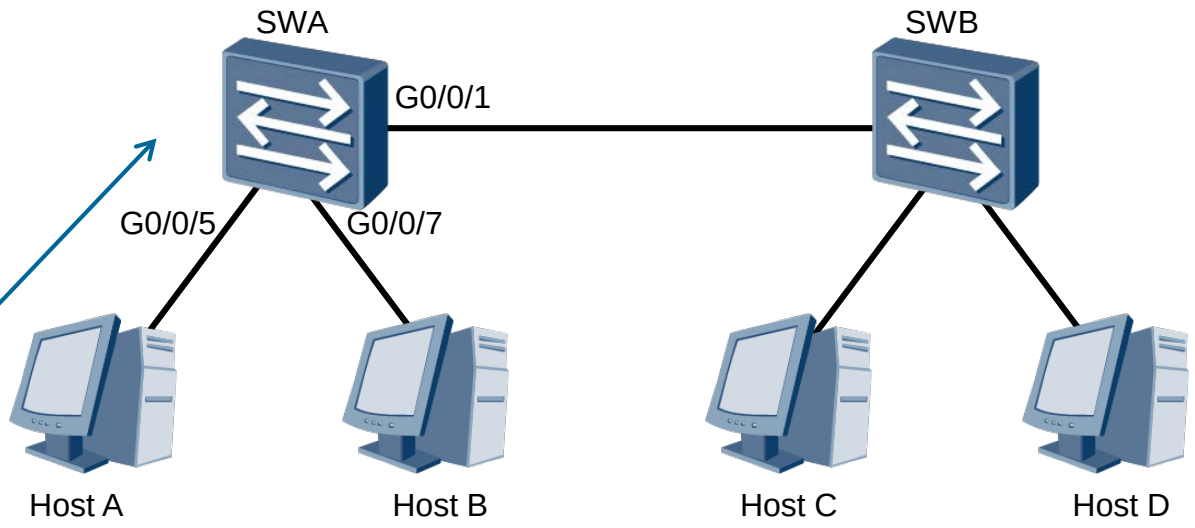
Setting the Port Link Type




```
[SWA]interface GigabitEthernet 0/0/1
[SWA-GigabitEthernet0/0/1]port link-type trunk
[SWA-GigabitEthernet0/0/1]interface GigabitEthernet 0/0/5
[SWA-GigabitEthernet0/0/5]port link-type access
```

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Assigning Ports to VLANs




```
[SWA]vlan 2
[SWA-vlan2]port GigabitEthernet 0/0/7
[SWA-vlan2]quit
[SWA-GigabitEthernet0/0/5]port link-type access
[SWA-GigabitEthernet0/0/5]port default vlan 3
```

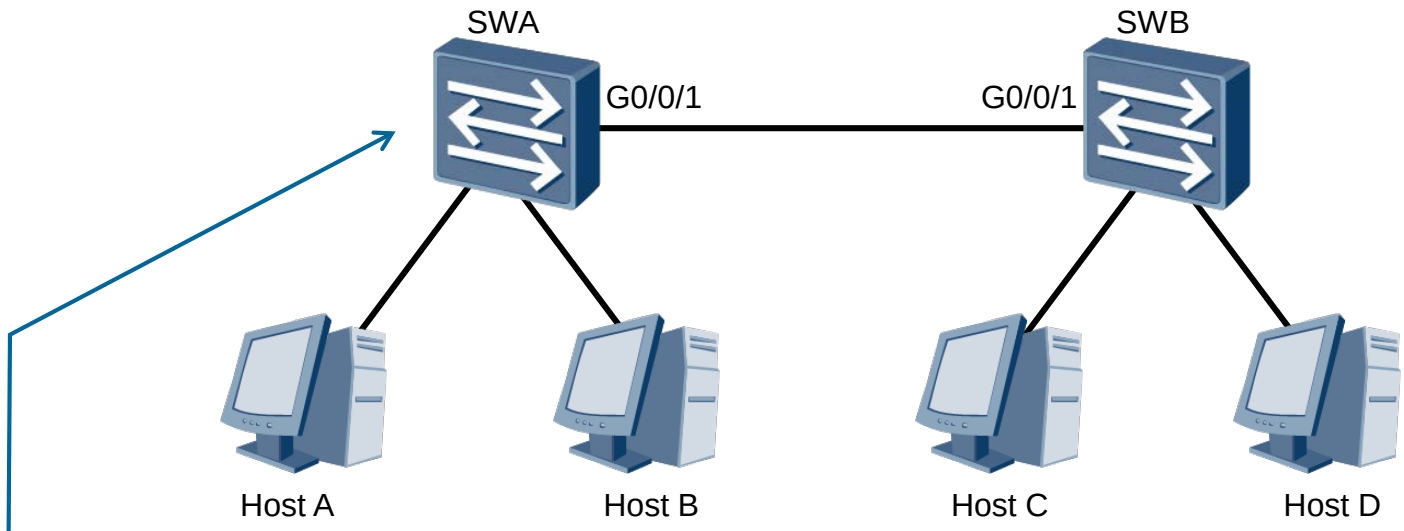
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Verifying VLAN Assignment

```
[SWA]display vlan
The total number of vlans is : 4
--------------------------------------------------------------
U:Up; D:Down; TG:Tagged; UT:Untagged; MP:Vlan-mapping;
ST:Vlan-stacking; #: ProtocolTransparent-vlan; *:Management-
vlan;
--------------------------------------------------------------
VID  Type    Ports
--------------------------------------------------------------
1    common  UT:GE0/0/1(U) ……
2    common  UT:GE0/0/7(D)
3    common  UT:GE0/0/5(U)
10   common
……
```

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Forwarding Over the Trunk




```
[SWA-GigabitEthernet0/0/1]port link-type trunk
[SWA-GigabitEthernet0/0/1]port trunk pvid vlan 10
[SWA-GigabitEthernet0/0/1]port trunk allow-pass vlan 2 3
```

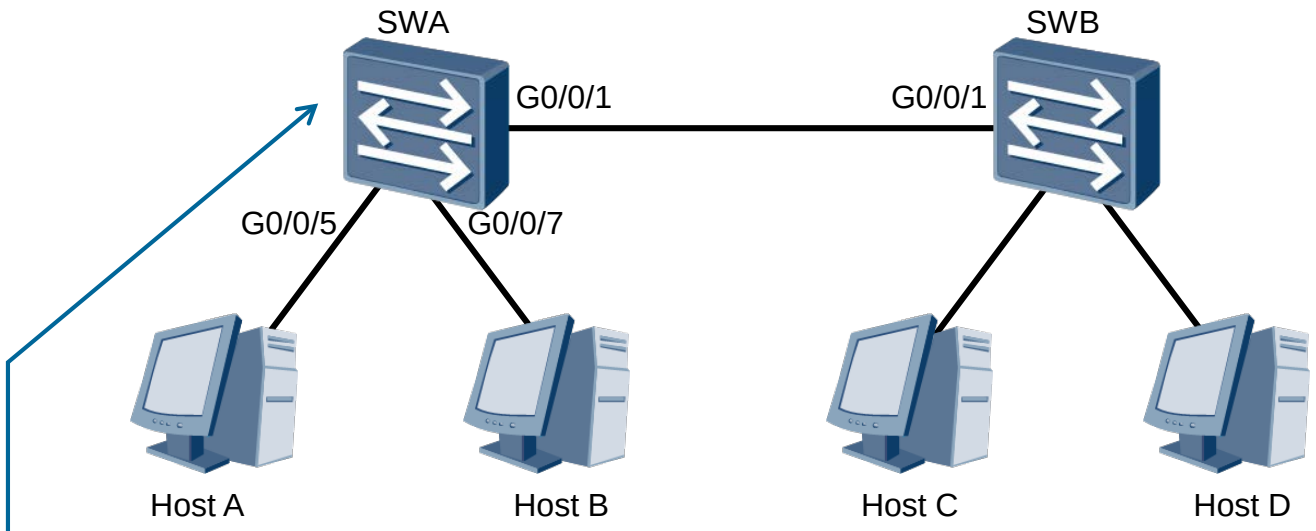
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Forwarding Over the Trunk

```
[SWA]display vlan
The total number of vlans is : 4
--------------------------------------------------------------
U:Up; D:Down; TG:Tagged; UT:Untagged; MP:Vlan-mapping;
ST:Vlan-stacking; #: ProtocolTransparent-vlan; *:Management-
vlan;
--------------------------------------------------------------
VID  Type    Ports
--------------------------------------------------------------
1    common  UT:GE0/0/1(U) ……
2    common  UT:GE0/0/7(D) TG:GE0/0/1(U)
3    common  UT:GE0/0/5(U) TG:GE0/0/1(U)
10   common
……
```

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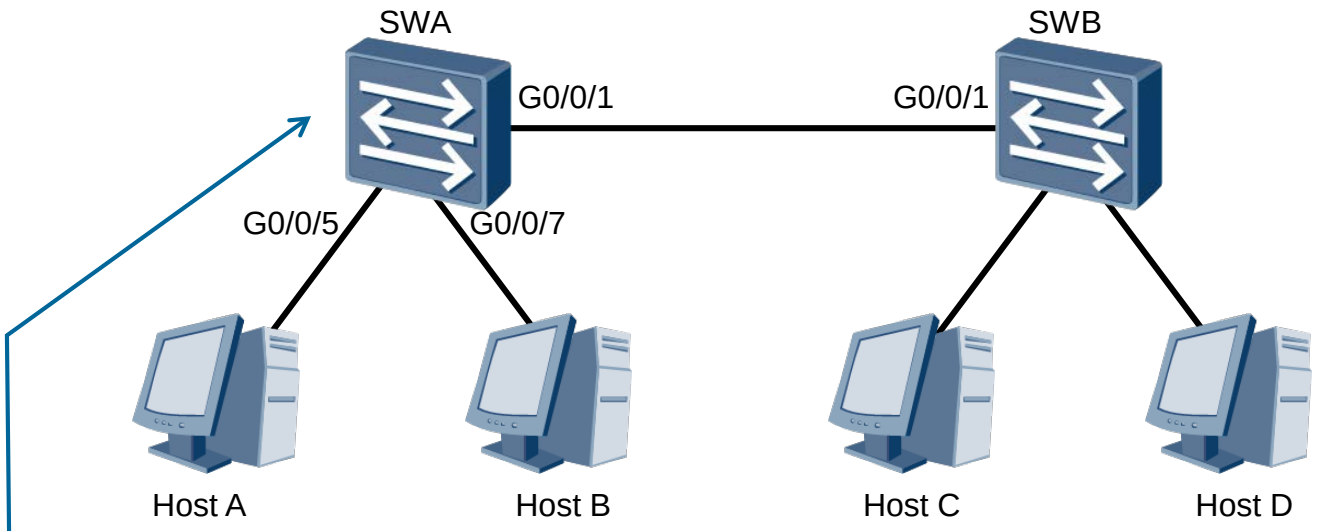
Configuring Hybrid Ports




```
[SWA-GigabitEthernet0/0/5]port link-type hybrid
[SWA-GigabitEthernet0/0/5]port hybrid pvid vlan 3
[SWA-GigabitEthernet0/0/5]port hybrid untagged vlan 3
[SWA-GigabitEthernet0/0/7]port link-type hybrid
[SWA-GigabitEthernet0/0/7]port hybrid pvid vlan 2
[SWA-GigabitEthernet0/0/7]port hybrid untagged vlan 2
```

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Configuring Hybrid Ports




```
[SWA-GigabitEthernet0/0/1]port link-type hybrid
[SWA-GigabitEthernet0/0/1]port hybrid tagged vlan 2 to 3
```

- Trunk links using the hybrid port link-type must enable tagging of VLAN frames before forwarding.

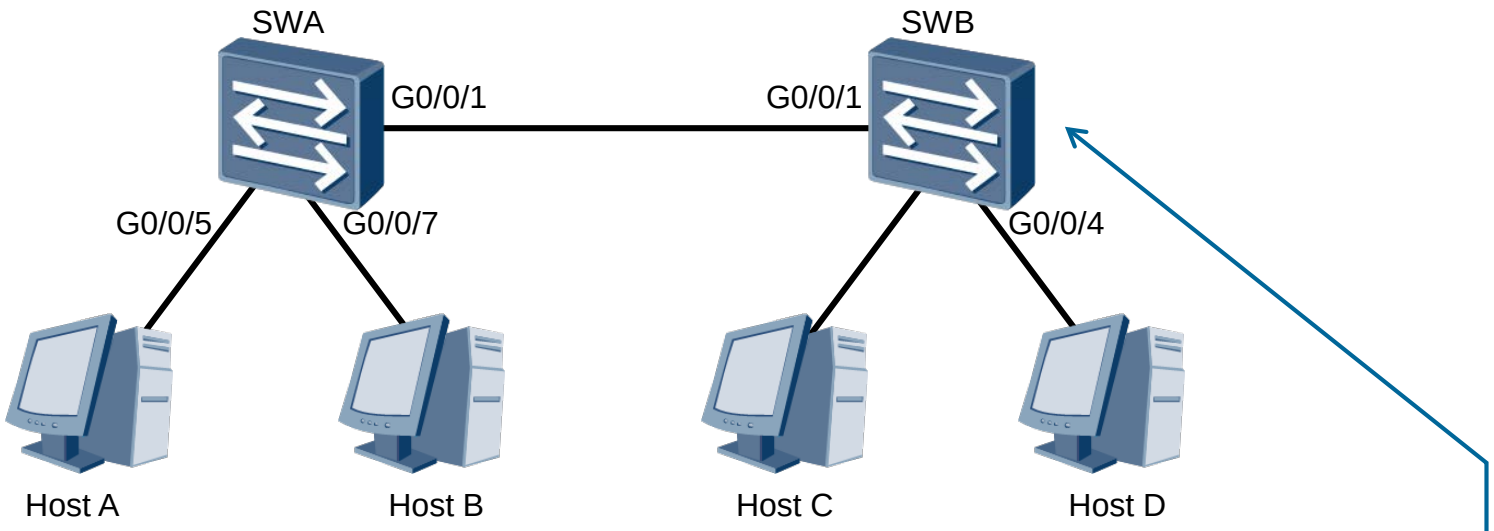
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Configuration Validation

```
[SWA]display vlan
The total number of vlans is : 4
--------------------------------------------------------------
U:Up; D:Down; TG:Tagged; UT:Untagged; MP:Vlan-mapping; ST:Vlan-
stacking; #: ProtocolTransparent-vlan; *:Management-vlan;
--------------------------------------------------------------
VID  Type    Ports
--------------------------------------------------------------
1    common  UT:GE0/0/1(U) ……
2    common  UT:GE0/0/7(D)
             TG:GE0/0/1(U)
3    common  UT:GE0/0/5(U)
             TG:GE0/0/1(U)
10   common
……
```

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Configuring Hybrid Ports




```
[SWB-GigabitEthernet0/0/4]port link-type hybrid
[SWB-GigabitEthernet0/0/4]port hybrid pvid vlan 3
[SWB-GigabitEthernet0/0/4]port hybrid untagged vlan 2 to 3
```

- Hybrid ports can be configured to receive VLAN traffic from multiple VLANs by simply removing the tag at the port interface.

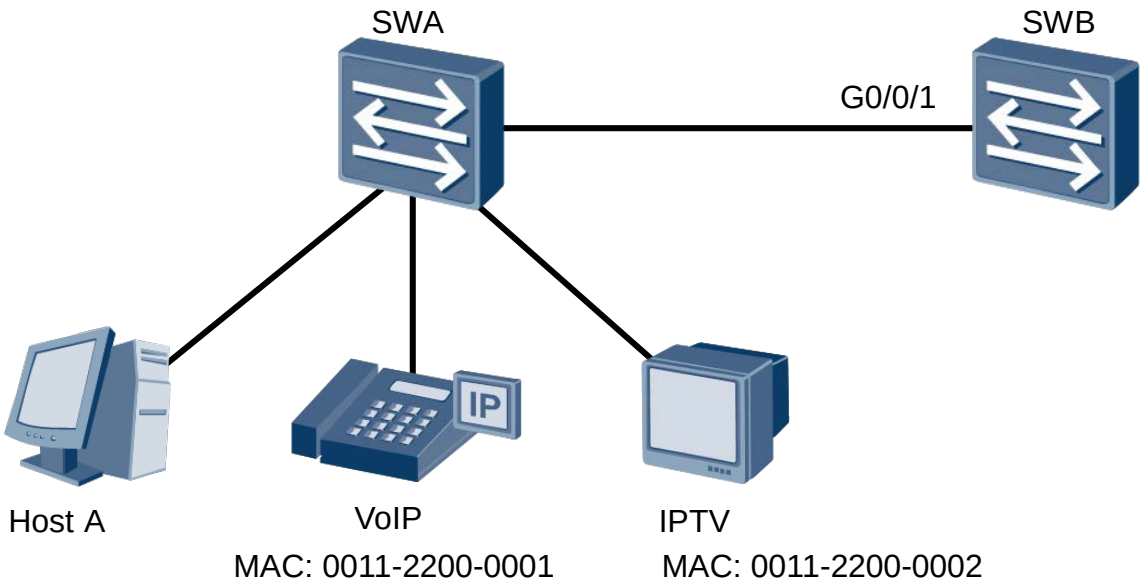
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Configuration Validation

```
[SWB]display vlan
The total number of vlans is : 3
--------------------------------------------------------------
U:Up; D:Down; TG:Tagged; UT:Untagged; MP:Vlan-mapping; ST:Vlan-
stacking; #: ProtocolTransparent-vlan; *:Management-vlan;
--------------------------------------------------------------
VID  Type    Ports
--------------------------------------------------------------
1    common  UT:GE0/0/1(U) ……
2    common  UT:GE0/0/4(U)
3    common  UT:GE0/0/4(U)
4   ……
```

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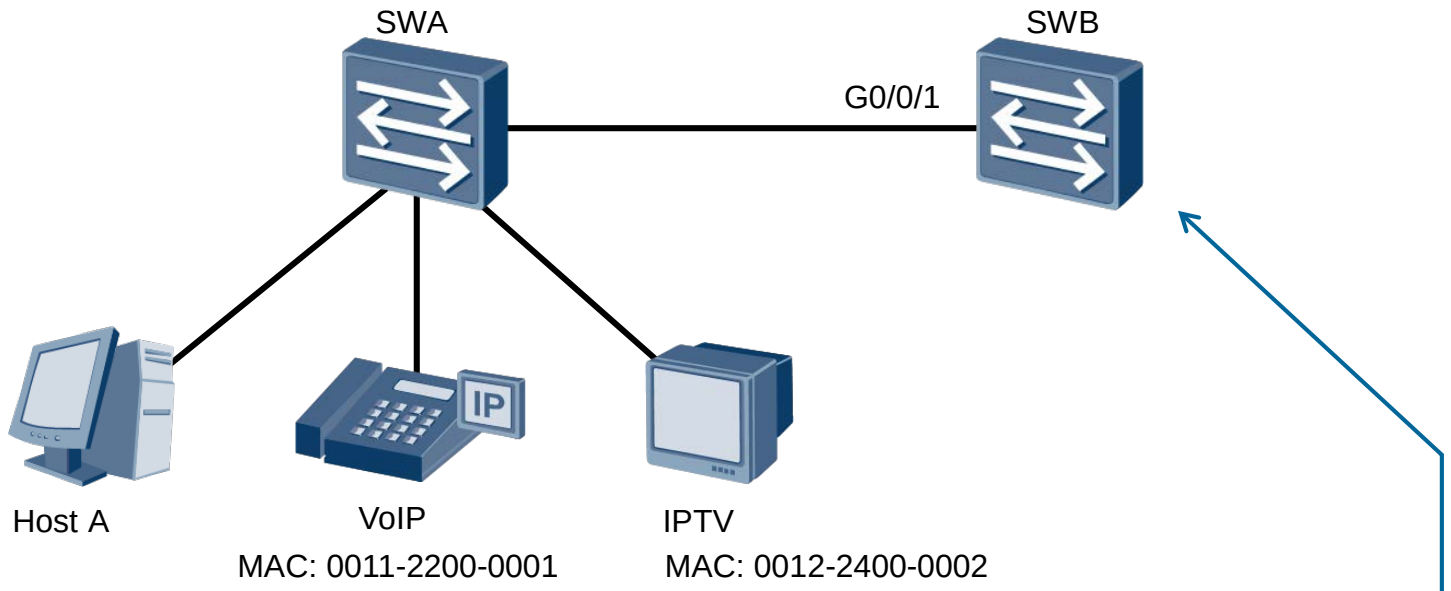
Voice VLAN Application



- Voice VLANs are used to distinguish, isolate and prioritize voice traffic over service traffic as a means of quality assurance.

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Voice VLAN Configuration




```
[SWB]vlan 2
[SWB-GigabitEthernet0/0/1]voice-vlan 2 enable
[SWB-GigabitEthernet0/0/1]voice-vlan mode auto
[SWB-GigabitEthernet0/0/1]quit
[SWB]voice-vlan mac-address 0011-2200-0000 mask ffff-ff00-0000
```

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Configuration Validation

```
[SWB]display voice-vlan status
Voice VLAN Configurations:
------------------------------------------------------------
Voice VLAN ID             : 2
Voice VLAN status         : Enable
Voice VLAN aging time     : 1440(minutes)
Voice VLAN 8021p remark   : 6
Voice VLAN dscp remark    : 46
------------------------------------------------------------
Port Information:
------------------------------------------------------------
Port                      Add-Mode  Security-Mode  Legacy
------------------------------------------------------------
GigabitEthernet0/0/1      Auto      Security       Disable
```

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Summary

- If a trunk link has a PVID of 5 and the command port trunk allow-pass vlan 2 3 is used, which VLAN traffic will be carried over the trunk?
- What action will be taken by an access port with a PVID of 2 when receiving an untagged frame?

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Thank you

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